

AKKOL, Kazakhstan

WMO: 350850

Lat: 52.000N

Lon: 70.950E

Elev: 1260

StdP: 14.04

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-21.3	-32.4	0.9	-26.9	-26.8	1.3	-21.1	26.7	11.9	23.1	17.1	3.8	0	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	88.0	63.5	84.5	62.9	81.3	61.9	66.9	80.9	65.6	78.9	64.2	76.8	7.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	88.4	71.3	61.0	83.9	70.1	59.4	79.3	69.3	32.2	80.5	31.1	79.2	30.0	77.0	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.1	18.7	16.5	DB	-32.4	93.5	5.1	3.2	-36.1	95.8	-39.0	97.7	-41.9	99.5	-45.6	101.8	(4)
			WB	-32.6	69.9	5.1	2.5	-36.3	71.7	-39.2	73.2	-42.1	74.6	-45.7	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	2.4	5.2	19.4	40.8	54.8	64.1	65.8	63.2	52.2	39.1	21.1	8.4
	DBStd	25.23	12.80	12.05	11.21	10.69	8.83	7.28	6.24	7.26	8.63	9.07	13.24	12.49
	HDD50	6627	1474	1253	948	308	53	4	0	3	74	352	867	1291
	HDD65	10637	1939	1673	1413	726	330	103	67	120	391	802	1317	1756
	CDD50	1717	0	0	0	32	202	425	490	413	140	15	0	0
	CDD65	252	0	0	0	0	14	74	91	65	8	0	0	0
	CDH74	3241	0	0	0	13	262	945	986	852	181	2	0	0
	CDH80	973	0	0	0	1	55	295	293	277	52	0	0	0
Wind	WSAvg	6.9	6.8	7.8	7.9	7.7	7.3	6.3	5.7	5.7	6.1	6.7	7.6	7.8
Precipitation	PrecAvg	14.6	0.8	0.7	0.9	1.0	1.3	1.5	2.7	1.5	0.9	1.2	1.1	1.1
	PrecMax	23.5	1.8	1.4	2.0	2.0	3.3	3.8	5.4	4.8	2.1	3.4	2.2	3.1
	PrecMin	8.6	0.2	0.2	0.2	0.3	0.1	0.2	0.5	0.5	0.1	0.2	0.6	0.2
	PrecStd	3.3	0.4	0.4	0.5	0.4	0.7	1.0	1.3	0.9	0.6	0.7	0.5	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	32.8	46.9	75.4	86.0	91.2	91.9	92.2	86.8	71.5	48.7	32.9
		MCWB	31.2	31.2	40.5	54.3	63.1	64.1	65.4	66.5	60.9	51.4	42.3	31.1
	2%	DB	28.8	29.6	39.6	69.1	80.3	86.9	86.8	87.0	79.2	64.5	43.6	30.9
		MCWB	27.7	28.1	36.0	52.6	58.1	62.5	64.6	63.7	57.2	49.0	39.9	29.4
	5%	DB	24.9	26.7	36.5	64.4	76.4	83.3	83.0	82.7	72.9	59.3	39.6	28.5
		MCWB	23.8	25.3	34.0	50.4	56.4	62.3	64.1	62.8	55.1	47.0	36.7	27.1
	10%	DB	21.4	23.4	33.9	58.8	72.7	79.5	79.4	78.6	68.1	53.8	36.0	25.4
		MCWB	20.2	22.0	31.7	47.4	55.2	61.2	63.3	61.5	53.1	44.8	33.9	24.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.3	31.5	40.9	56.9	63.0	67.8	69.0	68.4	62.7	53.5	44.1	31.7
		MCDB	32.2	32.4	47.1	70.9	78.5	83.8	82.3	82.3	82.0	65.3	46.1	32.3
	2%	WB	27.8	28.4	36.5	53.7	60.1	65.9	67.3	66.0	59.1	50.7	40.1	29.6
		MCDB	28.8	29.6	39.2	67.2	75.5	80.7	80.4	80.6	73.6	61.0	43.4	30.6
	5%	WB	23.9	25.3	34.2	50.9	58.1	64.2	66.0	64.4	56.5	47.9	36.8	27.2
		MCDB	24.8	26.6	36.3	62.0	72.6	78.0	78.7	78.1	70.3	57.5	39.3	28.3
	10%	WB	20.3	22.2	31.9	48.1	56.3	62.6	64.7	62.9	54.3	45.3	33.9	24.2
		MCDB	21.4	23.4	33.9	57.8	70.2	75.9	76.3	75.6	66.1	53.3	35.9	25.4
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.6	15.8	19.3	23.2	21.9	20.0	22.2	22.2	17.3	12.8	13.8
		MCDBR	14.1	14.2	13.4	28.8	29.8	28.5	27.3	29.2	31.2	28.3	14.9	14.3
		MCWBR	13.7	13.6	10.7	14.9	12.8	11.1	10.6	11.6	13.6	15.8	11.6	13.6
	5% WB	MCDBR	13.9	14.3	12.9	25.5	25.9	23.6	22.1	24.7	27.6	24.4	13.6	13.6
		MCWBR	13.7	13.9	10.8	14.5	12.7	10.9	10.0	11.7	13.0	14.7	11.1	13.3
Clear-Sky Solar Irradiance	taub	0.286	0.316	0.368	0.404	0.398	0.404	0.425	0.409	0.356	0.339	0.319	0.286	
	taud	2.066	2.026	1.971	2.140	2.235	2.288	2.224	2.250	2.352	2.302	2.144	2.050	
	Ebn at Noon	216	245	254	262	271	270	261	259	261	239	201	190	
	Edh at Noon	25	35	45	43	42	40	42	38	31	26	23	21	
All-Sky Solar Radiation	RadAvg	421	784	1233	1501	1855	1967	1804	1589	1181	642	309	283	
	RadStd	31	29	121	136	166	156	135	113	82	82	37	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page